

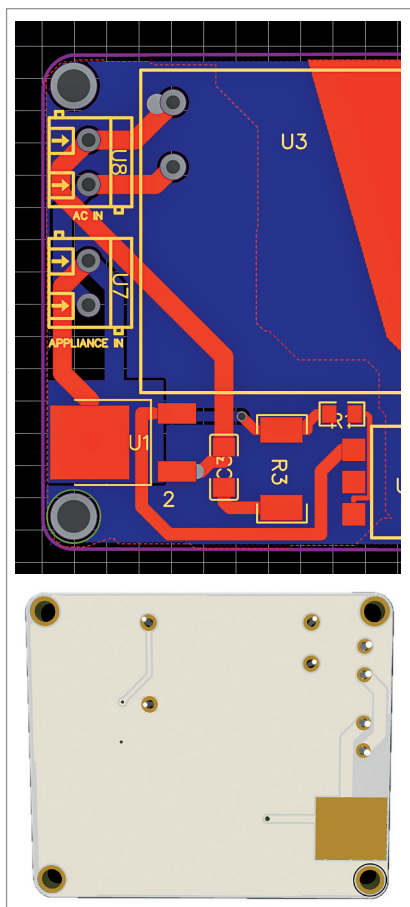


Fig. 7: Wider track width for AC; right solder pad provided for heatsink

to MT2 should be short and secure to reduce signal loss. In addition, placing a $0.1\mu\text{F}$ capacitor across VDD and GND near U2 helps filter electrical noise and enhances circuit stability. Fig. 6 shows the solid-state relay circuit (SSR), including the snubber circuit on the right-hand side.

Final combined circuit

The TTP223 touch signal drives the MOC3041SM optoisolator, which provides electrical isolation and triggers the BT136 to switch a 100-250V AC load. A snubber circuit protects the TRIAC from inductive spikes, ensuring reliable operation. The final system integrates the TTP223 touch sensor, BT136 TRIAC-based solid-state relay, and MOC3041SM on a compact $4\text{cm} \times 4\text{cm}$ PCB. The TTP223 Q pin connects to pin 1 of

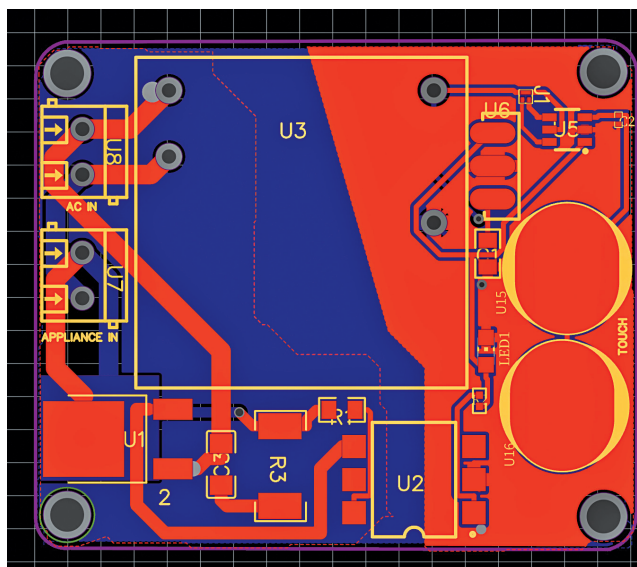


Fig. 8: Complete PCB (enlarged version)

the MOC3041SM through a 220Ω resistor (R2), with pin 2 connected to GND to drive the optocoupler's internal LED. Pin 4 of the MOC3041SM connects to the BT136 gate via a $1\text{ k}\Omega$ resistor (R1). In contrast, pin 6 connects to MT2, completing the triggering path for the TRIAC.

In the power switching section, MT1 of the BT136 is connected to the load through screw terminal U7, and MT2 is connected to the 220V live line. A snubber network ($100\Omega + 0.1\mu\text{F}$) is placed across MT1 and MT2 to protect against voltage spikes. The LD05-23B05R2 module (U8) provides a 5V output from a 100-250V AC input. Although not included in this design, a 1A, 250V fuse may be added for additional protection. A $0.1\mu\text{F}$ capacitor is placed near the 5V rail for noise decoupling. Power input is supplied

through the U8 screw terminal, and the AC load is connected via the U7 screw terminals.

PCB design

For PCB design, the layout is kept within a 4cm footprint to maintain a compact form factor. Terminal blocks are mounted on the top side of the PCB for AC input and load

wire connections. The AC-DC converter module is placed close to the terminal block for efficient routing. In the PCB layout, the tracks for AC connections are kept wide to handle high current and voltage, approximately 1.5mm in width, while the track width for the DC section is kept under 0.3mm , as lower current flows through it.

The TRIAC requires a large solder pad to act as a heat sink. This solder pad is placed on the bottom layer of the PCB and connected through vias for improved thermal dissipation. Two touch pads are provided on the opposite side of the AC terminal blocks to ensure user safety and prevent accidental contact with high-voltage lines. Four mounting holes are also provided at each corner of the PCB to enable secure mounting inside an enclosure or directly onto a wall.

Fig. 8 shows the wider track width for AC, and on the right-hand side, the a solder pad is provided for the heat sink. Fig. 9 shows the complete PCB of the solid-state relay. The final assembled PCB of the solid-state relay is shown in the prototype, in Fig. 1. **EFY**

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